1 Label-free representation learning

Input graph

Two views

Frozen $\bar{H}$

Graph encoder g_ψ (GRACE)

X, A

Contrastive loss

$Z^{(1)}$ $Z^{(2)}$

Projection head

After pretraining: $\bar{H} = \mathrm{RowNorm}(g_\psi(X, A))$

2 Granular-ball partitioning

Initial region

2-means split

Further splits

Final partition

Label-free splits; $K \le B_{\max} = 64$

Disjoint membership; circles show 2-D covers.

3 Consensus-weighted soft targets

(a) Selected ball

(b) Label proportions

(c) Soft-target construction

$p_{j,-v}$

Own label $\times(1-\alpha_{j,-v})$

Consensus $\times\alpha_{j,-v}$

Soft target t_v

Granular ball B_j

Exclude v's vote

$q_{j,-v} = (1-\beta)p_{j,-v} + \beta e_{m_{j,-v}}$

Ties retain the vote proportions.

Stronger label agreement gives more weight to consensus.

$t_v = (1-\alpha_{j,-v})e_{\tilde{y}_v} + \alpha_{j,-v}q_{j,-v}$

4 MLP training and graph-free inference

MLP training

Soft target t_v

MLP f_θ

$\hat{p}_v$

Soft cross-entropy

x_v

Training nodes $v \in \mathcal{V}_L$

Graph-free inference

$\hat{p}_u$

Class 3

x_u

Same f_θ

Observed training labels: C1 C2 C3 C4 C5 Non-voting node

Inference uses only node attributes.